Diabetes Regression Report - Tuning + Cross-Validation

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KNN regressor tuned with GridSearchCV and evaluated with 5-fold CV.

MODEL OVERVIEW

Name	KNeighborsRegressor
Type	Regression
Sklearn	1.8.0
Parameter Count	16

DATASET OVERVIEW

Features	10
Total Observations	331
CV Folds	5

HYPERPARAMETERS

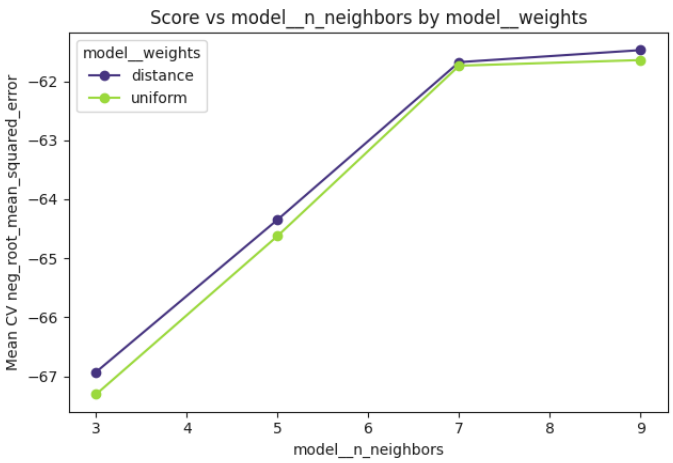
memory	None	transform_input	None
verbose	False	scaler	StandardScaler
model	KNeighborsRegressor	scaler__copy	True
scaler__with_mean	True	scaler__with_std	True
model__algorithm	auto	model__leaf_size	30
model__metric	minkowski	model__metric_params	None

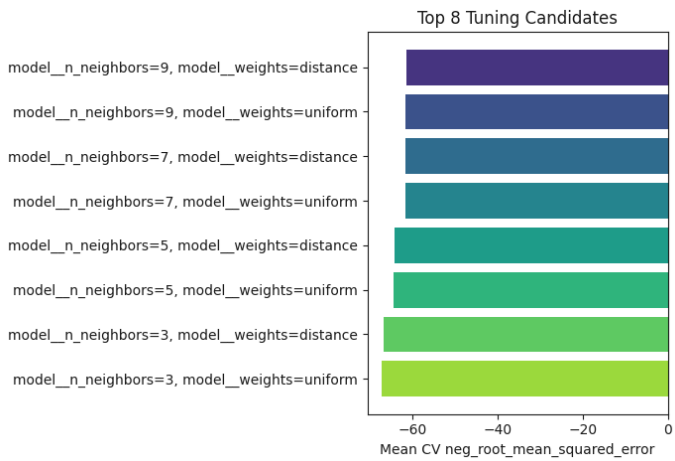
model__n_jobs	None	**model__n_neighbors**	9
model__p	2	**model__weights**	distance

Tuning Summary

Method	GridSearchCV
Metric	neg_root_mean_squared_error
CV Folds	5
Candidates	8
Best Score	-61.4718

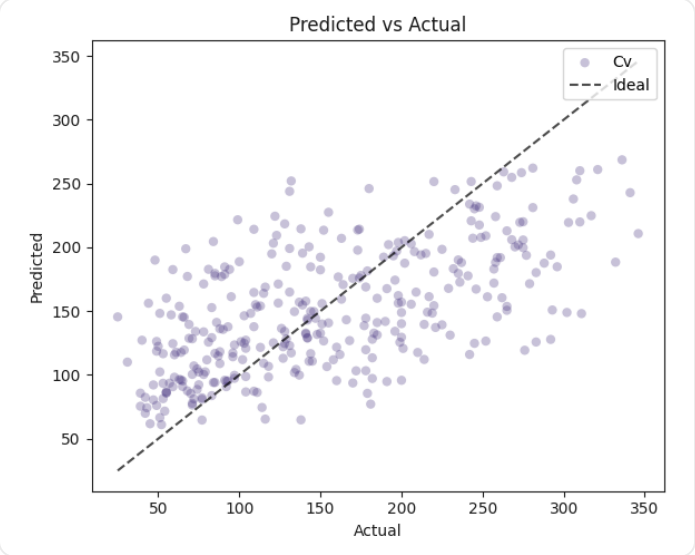
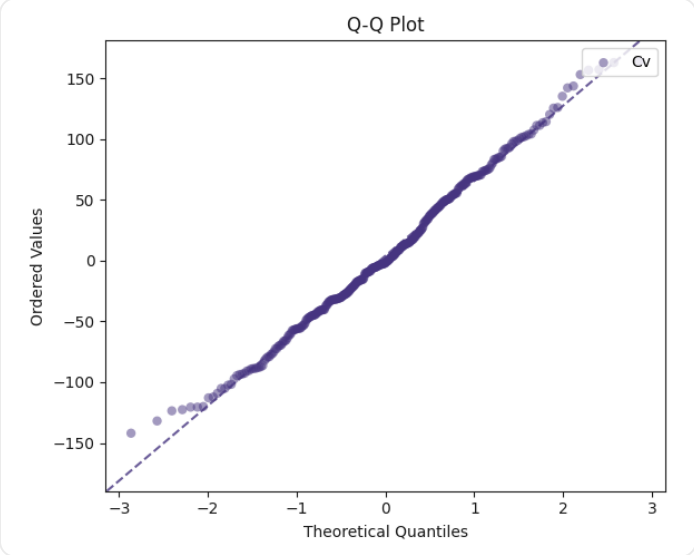
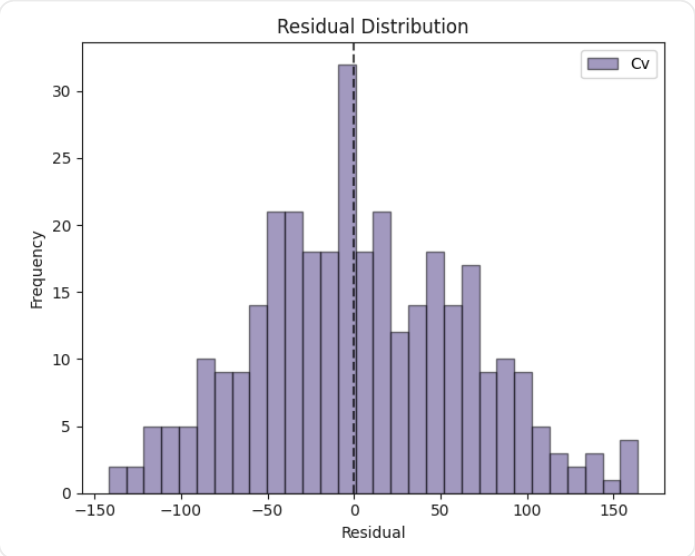
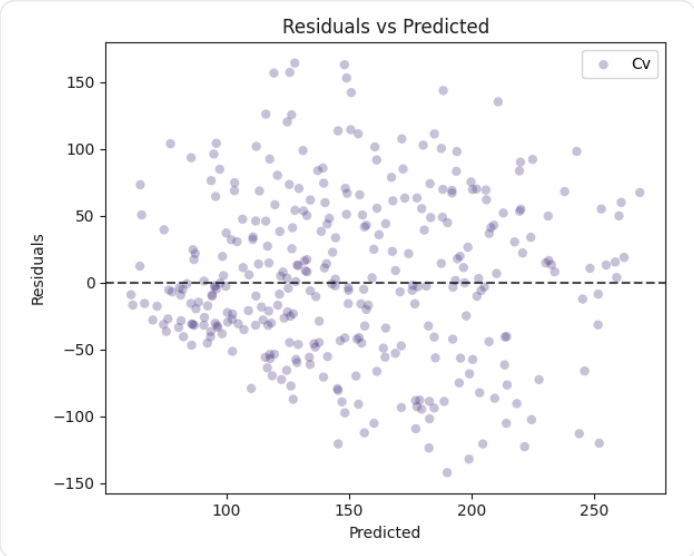
Tuning Plots





METRICS (CROSS-VALIDATION)

Metric	Mean	Std Dev	Min	Max
Mean Absolute Error	49.3585	3.0074	44.3420	52.6329
Max Error	155.8950	7.9642	142.1326	164.1776
Median Absolute Error	42.9693	5.3624	33.8980	49.0457
Mean Squared Error	3790.9658	417.7825	3045.2140	4163.9169

R² Score	0.3535	0.0975	0.1815	0.4759
Root Mean Squared Error	61.4718	3.4901	55.1835	64.5284

VISUALIZATIONS